

Fundamentals of Inference

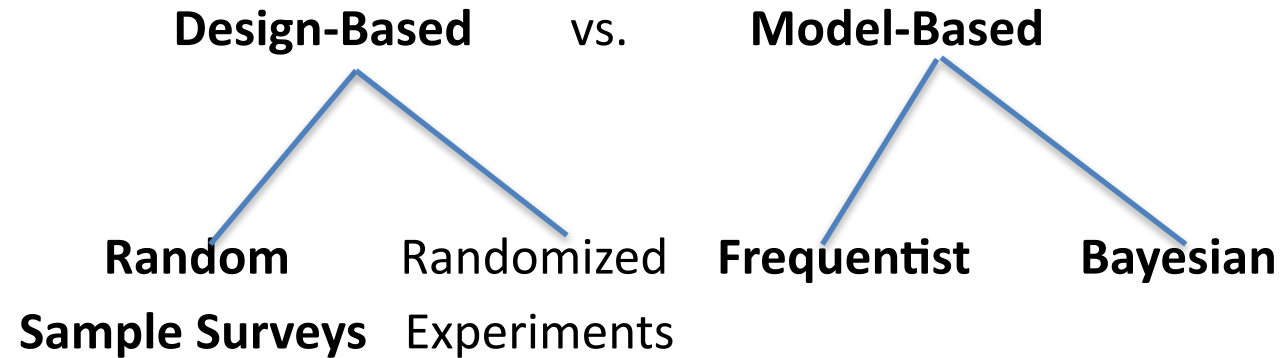
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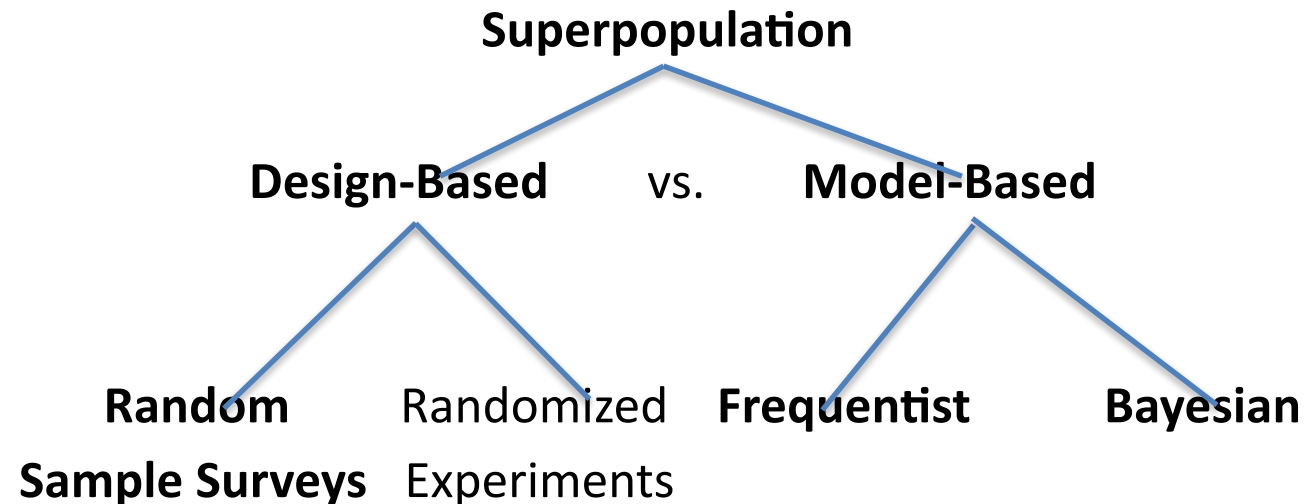
Module 9, Fall 2025

Topics in Inference

- **Methods for Making Finite Population Inference**



- **Add Superpopulation Model to provide Finite Population**



Horvitz-Thompson (HT) Estimator

- - Estimand of interest: **Population Total**, $Q(Y) = \sum_{j=1}^N Y_j$
 - Survey Design and Estimation Procedure, $q(y)$:

Survey design:

$$\Pr(I_j = 1) = \pi_j, j = 1, \dots, N$$

Sample

$$\{y_1, y_2, \dots, y_n\} \subset \{Y_1, Y_2, \dots, Y_N\}$$

HT estimator:

$$q(y) = \sum_{i=1}^n y_i / \pi_i = \sum_{j=1}^N I_j Y_j / \pi_j$$

Can write $q(y) = \sum_{i=1}^n w_i y_i$, $w_i = 1/\pi_i$ as a sample weighted total.

Horvitz-Thompson Estimator: Properties

- $q(y)$ is an unbiased estimator of $Q(Y)$ for any set of values of Y :

$$E[q(y)] = \sum_{j=1}^N E(I_j) Y_j / \pi_j = \sum_{j=1}^N Y_j = Q(Y)$$

(note: $E(I_j) = \text{Pr}(I_j = 1) = \pi_j$)

- A related estimator, $q(y)$, of the population mean $Q(Y) = \frac{1}{N} \sum_{j=1}^N Y_j$:

$$q(y) = \sum_{i=1}^n (y_i / \pi_i) / \sum_{i=1}^n (1 / \pi_i).$$

If $\sum_{i=1}^n 1 / \pi_i = N$, then $E[q(y)] = Q(Y)$, i.e., $q(y)$ is an unbiased estimator of $Q(Y)$.

Often only $E[\sum_{i=1}^n 1 / \pi_i] = E[\sum_{j=1}^N I_j / \pi_j] = N$ then $q(y)$ is a ratio estimator of $Q(Y)$.

So $E[q(y)] \neq Q(Y)$, i.e., is biased, but $q(y)$ is a **consistent** (see Module 7 slide 46) estimator of $Q(Y)$.

Unbiased on average, but...

- An estimator can be unbiased, i.e., averaged across all samples
- But we only select one sample....
- What if no sample is “good” by itself?

Basu's Elephant: Circus Manager vs. Statistician

- A circus has a herd of 11 elephants of varying sizes
 - Given the food needed when going from one town to another, the manager needs an estimate of the total weight of the herd
- But only one elephant can be weighed!
- For years, the circus manager takes the following approach:
 - Pick out one elephant that he considers to be the “middle-of-the-herd” in terms of weight
 - Weigh that elephant and multiply by 11
- He has not been sure about his methodology and somebody told him that what he is doing is not a probability sample

Basu's Elephant, continued

- Circus manager meets a statistician and describes his method
- Statistician sees some value in the methodology but suggests a modification to make it a probability sampling design
- Proposed design:
 - Choose the the middle of the herd elephant with probability 0.9
 - Assign the selection probability of 0.01 to each of the remaining 10 elephants
 - Use Horvitz-Thompson procedure for estimating the total
- Circus manager then decides to implement this method but soon got into big trouble

Basu's Elephant: Trouble!

- Suppose that the 11 elephants' weights (in pounds) are {9000, 9500, 10000, 10100, 10500, 11000, 11100, 11600, 11700, 12800, 13500}: **sum=120,800**, mean=10,982
- Statistician's procedure:
 - If the selected value is 11,000: the estimate of the TOTAL weight is $11,000/0.9=12,222$!!!
 - If the selected value is 13,500: the estimate of the total weight is $13,500/0.01=1,350,000$!!!!
 - If the selected value is 9,000: the estimate of the total is $9,000/0.01=900,000$!!!
 - **Every estimate is problematic but the average (weighted by the probability of the sample) is the true value**
- Previous estimate by circus manager: **$11,000 \times 11=121,000$**
- Circus fired the statistician, gave a bonus to the manager
- Basu jokes that now the statistician decides to become a professor
- **Reference:** Basu, D. (1971). An essay on the logical foundations of survey sampling, Part 1 (with discussion), In: Godambe and Sprott (Eds.), Foundations of Statistical Inference, 203–242, Holt, Reinhart and Winston, Toronto. pp. 212-213.

Model-Based Inference

- Basu's elephant example is an extreme case of where the randomization (design-based) approach to inference can lead to poor conclusions based on a single sample.
- We consider an overview of **model-based inference**
- What is a model in this context?
- We will first talk about Frequentist inference and then Bayesian inference that are both model-based

Statistical Models

- A **statistical model** is a [mathematical model](#) that embodies a set of [statistical assumptions](#) concerning the generation of [sample data](#) (and similar data from a larger [population](#)). A statistical model represents, often in considerably idealized form, the **data-generating process**. (ref. Wikipedia)
 - Think of a machine, governed by parameters (regression coefficients, variances, etc.), that continuously spits out data!
- A **statistical model** is usually specified as a mathematical relationship between one or more [random variables](#) and other non-random variables.

Statistical Models

“All models are wrong, but some models are useful.” -- Box

Two Major Issues in Estimation and Inferential for Analyzing Complex Survey Data

- Sample weighting
 - Use of **sample design weights vs. not use of sample design weights** (unweighted) in estimation and inference
- Variance estimation
 - Use of **design-based vs. model-based variance** estimation

Design-based variance estimation take account of the sample design, e.g., sample weights, any stratification or clustering, in the variance estimation without strong model assumptions, e.g., uses Taylor linearization method and replication methods like jackknife or bootstrap to estimate the variances.

Model-based variance estimation may not fully take account of the sample design and can make model assumptions, e.g., normal distribution of the data.

Design weighted vs. Unweighted estimation of means

- Design-based weighted estimators give approximately unbiased estimates of population parameters.
Why not always do weighted estimation?

One reason weighted estimation can be inefficient. Consider

$$\bar{y}_w = \frac{\sum_{i=1}^n w_i y_i}{\sum_{i=1}^n w_i} \text{ and } \bar{y}_u = \frac{\sum_{i=1}^n y_i}{n}$$

- Consider a probabilistic model:

Let sample weights w_i be fixed constants and y_i are iid (independent identically distributed) random variables with mean μ variance σ^2 (similar to the superpopulation setup in Module 7 slide 49) then $\bar{y}_w$ and $\bar{y}_u$ are unbiased estimators for μ , i.e.,

$$E[\bar{y}_w] = E\left[\frac{\sum_{i=1}^n w_i y_i}{\sum_{i=1}^n w_i}\right] = \frac{\sum_{i=1}^n w_i E[y_i]}{\sum_{i=1}^n w_i} = \frac{\sum_{i=1}^n w_i \mu}{\sum_{i=1}^n w_i} = \mu$$

$$\text{and } E[\bar{y}_u] = E\left[\frac{\sum_{i=1}^n y_i}{n}\right] = \frac{\sum_{i=1}^n E[y_i]}{n} = \frac{\sum_{i=1}^n \mu}{n} = \mu$$

Note w_i are unrelated to y_i . It can be shown that

$$\text{Var}(\bar{y}_w) = \sigma^2 \sum_{i=1}^n w_i^2 / (\sum_{i=1}^n w_i)^2$$

$$\text{Var}(\bar{y}_u) = \sigma^2 / n.$$

Module 7 slide 49

- Let the target finite population values be $\{Y_1, Y_2, \dots, Y_N\}$ be generated from a superpopulation model:
 Y_i are independent and identically distributed random variables with mean μ and variance σ^2 .

Design weights vs. unweighted for estimation of means

- How much smaller is $Var\bar{y}_u$ than $Var\bar{y}_w$? (Assume y_i are iid random variables with mean μ variance σ^2)

$$\begin{aligned} inefficiency_0 &= \frac{Var\bar{y}_w - Var\bar{y}_u}{Var\bar{y}_w} = 1 - \frac{Var\bar{y}_u}{Var\bar{y}_w} \\ &= 1 - \frac{(\sum_{i=1}^n w_i)^2}{n \sum_{i=1}^n w_i^2} \approx \frac{CV^2}{1+CV^2} \end{aligned}$$

$CV = \frac{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (w_i - \bar{w})^2}}{\frac{1}{n} \sum_{i=1}^n w_i}$ the coefficient of variation of sample weights, i.e., the standard deviation of the weights divided by mean of the weights.

- $Inefficiency_0$ is the proportional decrease in the variance by using $\bar{y}_u$ instead of $\bar{y}_w$.
How does the $inefficiency_0$ above relate to $def f_{weight}$ defined on slide 20 of Module 8?
Recall that the $def f_{weight} = 1 + CV^2$ then $inefficiency_0 = 1 - (def f_{weight})^{-1}$

The subscript "0" in " $Inefficiency_0$ " refers to the computation when $\bar{y}_u$ is unbiased for the population mean, i.e., w_i are unnecessary or called "noninformative".

Table shows *Inefficiency*₀ varies by survey and subpopulation

Table 4.4-2. Inefficiency of Using Sample Weights when Unnecessary (Formula (4.4-3)) for Some Data Sets Analyzed in This Book

Survey ^a	Restrictions	Sample Size	Coefficient of Variation of the Weights	Inefficiency ₀
NHANES III	(a) Men ages 50+ years	3092	115%	57%
	(b) Medical examination			
1988 NMIHS	Live births	9953	98%	49%
NHANES I	(a) Women ages 25–74 years	5270	97%	48%
Epidemiologic Followup Study	(b) 1971–74 NHANES I component			
NHANES II	Boys ages 2–18 years	3667	71%	34%
1986 NMFS	Digestive cancer deaths	785	60%	26%
1987 NHIS	(a) Men ages 18+ years	16,008	53%	22%
	(b) Cancer control or epidemiology supplements			
NHANES II	White men ages 40–59 years	595	41%	14%
HHANES	Ages 2–11 years	2012	24%	5%
LSOA (baseline weights)	Women	4667	21%	4%

^aBrief descriptions of these surveys are given in Appendix A

Definition of acronyms

NHANES = National Health and Nutrition Examination Survey

NMIHS = National Maternal and Infant Health Survey

NMFS = National Mortality Followback Health Survey

NHIS = National Health Interview Survey

HHANES = Hispanic Health and Nutrition Examination Survey

LSOA = Longitudinal Study on Aging

Design weights vs. unweighted for estimation of means

- $Inefficiency_0 = 1 - \frac{Var(\bar{y}_u)}{Var(\bar{y}_w)} \approx \frac{CV^2}{CV^2+1}$ may not be accurate:
- Does not account for the actual sample design and adjustments to weights.
Stratification and poststratification can reduce variance of $\bar{y}_w$.
Additionally, sampling may lead to correlation between y_i and w_i , e.g., if the selection probabilities are $\approx$ proportional to the y_i and/or y_i may be correlated if there is cluster sampling (ρ).
- Only accounts for analyses of single means, e.g., not differences in means or regression coefficients.
- General expression for one dimensional parameter

$$Inefficiency_0 = 1 - \frac{Var(\hat{\theta}_u)}{Var(\hat{\theta}_w)}$$

estimated by

$$Inefficiency_0 = 1 - \frac{\widehat{var}(\hat{\theta}_u)}{\widehat{var}(\hat{\theta}_w)}$$

" $\widehat{var}$ " is an estimated variance incorporating the sample design, e.g., using Taylor linearization or a replication such as jackknife variance estimation.

Modeling sample weights, an alternative to sample weighted (HT) estimation: A frequentist approach

- Estimating a Population Mean
- Estimating a Population Mean Difference
- Estimating a General Parameter – Linear and Logistic Regression

Modeling sample weights, an alternative to sample weighted (HT) estimation: A frequentist approach

- Estimating a Population Mean
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Estimating a Population Mean

- When inefficiency of sample weighting (based on $Inefficiency_0$) is unacceptably large, one might consider alternative approaches.
- One approach is augmenting the analysis model with **additional variables** used for the survey design and then use unweighted estimation.
- Consider we want to estimate a population mean $\bar{Y} = \frac{1}{N} \sum_{i=1}^N Y_i$ for a variable Y using a stratified random sample of size n with sex as the sampling strata, where there is **over sampling** of women, i.e.,

$$n_{women}/N_{women} > n_{men}/N_{men}, \text{ where } N = N_{men} + N_{women}, n = n_{women} + n_{men}.$$

- Suppose $Inefficiency_0$ of $\bar{y}_w$ relative to $\bar{y}_u$ is mostly from the oversampling of the women.

Estimating a Population Mean (Cont.)

- Consider first estimating stratified population $\bar{Y}_{women}$ and $\bar{Y}_{men}$ by $\bar{y}_{w,women}$ and $\bar{y}_{w,men}$, each of these two sex specific sample means have little inefficiency. But we want to estimate $\bar{Y}$. One way is putting these two estimators $\bar{y}_{w,women}$, $\bar{y}_{w,men}$ together as

$$\bar{y}_{model} = (N_{women}\bar{y}_{w,women} + N_{men}\bar{y}_{w,men}) / (N_{men} + N_{women}),$$

$\bar{y}_{model}$ is an unbiased estimator for $\bar{Y}$ (assume that the N_{men} , N_{women} are known) because $\bar{y}_{w,women}$, $\bar{y}_{w,men}$ are unbiased for $\bar{Y}_{women}$ and $\bar{Y}_{men}$. This is modeling the mean using the variable sex. Is $\bar{y}_{model}$ more efficient than $\bar{y}_w$? Answer is No.

- BTW you can use [sample weighted linear regression](#) modeling to obtain $\bar{y}_{w,women}$, $\bar{y}_{w,men}$:

$$y_i = \alpha + \gamma s_i + e_i$$

$s_i = 1$ or 0 for women or men, respectively. Then $\bar{y}_{w,women} = \hat{\alpha}_w + \hat{\gamma}_w$ and $\bar{y}_{w,men} = \hat{\alpha}_w$.

Modeling survey design sample weighting, an alternative to sample weighted (HT) estimation: A frequentist approach

- Estimating a Population Mean
- Estimating a Population Mean Difference
- Estimating a General Parameter – Linear and Logistic Regression

Estimating a Population Mean Difference

- Controlling for sample design variable (e.g., sex) is more complicated but potentially more useful for estimating association between variables.
- E.g., consider estimating population mean difference of a continuous outcome Y for individuals exposed to high (HI) and low (LO) levels of risk factor,

$$DIF = \bar{Y}_{HI} - \bar{Y}_{LO}$$

with weighted mean estimated difference

$$dif = \bar{y}_{HI} - \bar{y}_{LO}$$

- Suppose dif is highly inefficient estimator of DIF due to highly variable sample weights (i.e., large CV of w'_i s for the HI, LO). If a large part of the variability is from oversampling women the sex-specific mean differences

$$\begin{aligned} dif_{men} &= \bar{y}_{w,HI,men} - \bar{y}_{w,LO,men} \\ dif_{women} &= \bar{y}_{w,HI,women} - \bar{y}_{w,LO,women} \end{aligned}$$

will not be so inefficient.

Estimating a Population Mean Difference (cont.)

- There may be substantive reasons why dif_{men} , dif_{women} are of equal or of more importance than dif . Note that these sex-specific differences will have lower inefficiency.
- Returning to estimating DIF , there are two choices for the pop. parameter for the overall association of risk factor and Y

$$DIF = \bar{Y}_{HI} - \bar{Y}_{LO}$$

and

$$DIF_{Strat} = \frac{N_{men}DIF_{men} + N_{women}DIF_{women}}{N_{men} + N_{women}}$$

- DIF and DIF_{Strat} can be quite different even when $DIF_{men} = DIF_{women}$; see Table 4.5-1 where $DIF_{Strat} = 0.2$ but $DIF = 0.34$. Why? Sex is related to High-risk vs. Low-risk membership and sex is related to population mean of Y (higher mean values of Y in Men than Women)

Estimating a Population Mean Difference (cont.)

Table 4.5-1. Population Means (Population Sizes) for a Hypothetical Population by Sex and Risk Group

	Men	Women	Total
Group 1 (high risk)	.8 (2×10^6)	.4 (10^6)	.67 (3×10^6)
Group 2 (low risk)	.6 (10^6)	.2 (2×10^6)	.33 (3×10^6)
Group difference	.2	.2	.34

- 2 questions:
 - (1) When is it appropriate to estimate DIF_{Strat} instead of DIF ? E.g., if the objective is to change people's risk factor status from high to low then DIF_{Strat} gets closer to the expected difference within individuals.
 - (2) Are estimators of DIF_{Strat} as variable as estimators of DIF ? This will depend on how DIF_{Strat} is estimated.

Estimating a Population Mean Difference (cont.)

- A natural estimator of DIF_{Strat} is

$$dif_{Strat} = \frac{N_{men}dif_{men} + N_{women}dif_{women}}{N_{men} + N_{women}}$$

will tend to be as variable as $dif = \bar{y}_{HI} - \bar{y}_{LO}$.

- Instead of dif_{Strat} , we consider using linear regression modeling that includes sex as a covariate:

$$y_i = \alpha + \beta r_i + \gamma s_i + e_i$$

where for each sample person the binary variables $r = 0$ or 1 according to low or high risk factor levels, $s = 0$ or 1 according to sex and e is the error. Estimate the parameters α, β, γ using unweighted (sample weights are all 1) ordinary-least-squares (OLS) estimation, then $\hat{\beta}_u$ is an estimate of the difference in HI and LO levels of risk factor adjusting the sampling strata of sex by including sex in the regression model.

- If $DIF_{men} = DIF_{women}$, i.e., **no r by s interaction in the finite population**, then $\hat{\beta}_u$ will be approximately unbiased for DIF_{Strat} . If most of variability of the weights are due to over sampling women then $\hat{\beta}_u$ will be more efficient than the weighted estimator of β .

Summary - Estimating a Population Mean Difference

- If it can be substantively justified to control for a variable in an analysis, e.g., sex, then estimate risk/outcome differences separately by that variable.
- If an overall measure of association is of interest, e.g., mean difference of Y for HI vs. LO risk factor is required, and if risk factor-by-sample design variables interactions are small then using an unweighted regression model including the sample design variables as main effects in the regression model, e.g., the sample design stratification variable sex as a (main effect) covariate, can result in more efficient estimation of risk factor/outcome association.

Modeling survey design sample weighting, an alternative to sample weighted (HT) estimation: A frequentist approach

- Estimating a Population Mean
- Estimating a Population Mean Difference
- Estimating a General Parameter – e. g., coefficients for linear or logistic regression

A data example modeling survey design to estimate an association

- What is the association between maternal smoking and education?
- The data from the 1988 National Maternal and Infant Health Survey (NMIHS)
- Logistic regression modeling

1988 National Maternal and Infant Health Survey (NMIHS)

- Use the live birth component of the 1988 NMIHS. Stratified random sample of 13,415 birth certificates from a list frame of about 3.9 million live births.
- Higher sampling rates (oversampling) for black vs. nonblack babies and low-birthweight vs. normal birthweight babies .
- Mothers were queried (mail or phone or face-to-face) to obtain information not on the birth certificates, e.g., maternal smoking, 9,953 responded.
- Nonresponse adjustments to original sample weights were based on birthweight, race, marital status and maternal age.
- Poststratification adjustments of weights to vital statistics based on same variables as nonresponse adjustments.

Data example modeling survey design to estimate an association in NMIHS

Table 4.2-1. Sampling Strata and Sampling Rates of 1988 NMIHS

Stratum Number	Strata		Sampling Rate	Base Sample Weight
	Race	Birthweight (grams)		
Black				
1		< 1500	1/14	14
2		1500–2499	1/55	55
3		≥ 2500	1/113	113
Nonblack				
4		< 1500	1/29	29
5		1500–2499	1/160	160
6		≥ 2500	1/720	720

Data example modeling survey design to estimate an association in NMIHS

Table 4.2-2. Percent Distribution of Live Births by Age of Mother, United States, 1988

Age of Mother (Years) (1)	Vital Statistics ^a (2)	NMIHS-Sampled Birth Certificates (<i>n</i> = 13,415)		NMIHS-Interviewed Mothers Only (<i>n</i> = 9953)		
		Weighted (3)	Unweighted (4)	Fully Weighted ^b (5)	Partially Weighted ^c (6)	Partially Weighted ^d (7)
15–19	12.3	12.3	17.5	12.3	12.2	10.9
20–24	27.4	27.6	29.9	27.4	27.6	26.4
25–29	31.8	31.5	28.0	31.7	31.5	32.5
30–34	20.6	20.6	17.6	20.6	20.6	22.0
35–39	6.9	6.9	6.1	6.9	6.9	7.1
40–44	1.0	1.0	0.9	1.1	1.1	1.1
45–49	0.0	0.0	0.0	0.0	0.0	0.0

^aBased on *N* = 3,898,922 birth certificates; 10,588 births with mothers under the age of 15 have been excluded to be consistent with the sample frame of 1988 NMIHS. Data are from NCHS (1990, p. 97).

^bSample weights incorporate differential probabilities of selection, nonresponse and poststratification adjustments.

^cSample weights only incorporate differential probabilities of selection, and nonresponse adjustments.

^dSample weights only incorporate differential probabilities of selection.

Data example modeling survey design to estimate an association in NMIHS

- Using logistic regression to study education association with maternal smoking:

$$\log\left(\frac{\pi}{1-\pi}\right) = \alpha + \beta_{race}I(Black) + \beta_{ed1}I(educ < 12) + \beta_{ed2}I(educ > 12)$$

π is probability that mother smoked and $I(\cdot)$ are dummy indicator variables, $ed1$ and $ed2$ for $educ < 12$ and $educ > 12$. Note birthweight was not included to avoid over-adjustment since smoking reduces birthweight. However, marital status and maternal age could have been included.

Table 4.3-4. Logistic Regression Analysis of the Probability of Smoking on Race and Mother's Education

Independent Variable	Weighted Coefficient $\pm$ SE	Unweighted Coefficient $\pm$ SE
Intercept	-.45 $\pm$.06	-.52 $\pm$.05
Race (black vs. nonblack)	-.33 $\pm$.05	-.33 $\pm$.04
Mother's education < 12 years vs. 12 years	-.26 $\pm$.07	-.03 $\pm$.05
Mother's education > 12 years vs. 12 years	-.88 $\pm$.08	-.63 $\pm$.06

Data from the 1988 NMIHS.

Data example modeling survey design to estimate an association in NMIHS

- Using logistic regression:

$$\log\left(\frac{\pi}{1-\pi}\right) = \alpha + \beta_{race}I(Black) + \beta_{ed1}I(educ < 12) + \beta_{ed2}I(educ > 12) + \beta_{race \times ed1}I(Black, ed1) + \beta_{race \times ed2}I(Black, ed2)$$

where π is probability maternal smoking.

Table 4.3-5. Logistic Regression Analysis of the Probability of Smoking on Mother's Education, Separately for Blacks and Nonblacks

Independent Variable	Blacks		Nonblacks	
	Weighted Coefficient ± SE	Unweighted Coefficient ± SE	Weighted Coefficient ± SE	Unweighted Coefficient ± SE
Intercept	-1.12 ± .06	-1.03 ± .05	-.38 ± .06	-.30 ± .05
Mother's education < 12 years vs. 12 years	.24 ± .08	.24 ± .07	-.37 ± .08	-.33 ± .07
Mother's education > 12 years vs. 12 years	-.23 ± .10	-.26 ± .09	-.99 ± .10	-.97 ± .08

Data from the 1988 NMIHS.

Modeling survey design sample weighting: Estimating a general parameter

- Begin with a model for the analysis if you had a simple random sample, i.e., unweighted.
- Check the inefficiency of the estimator $\hat{\theta}$ of interest

$$Inefficiency_0 = 1 - \frac{\widehat{var}(\hat{\theta}_u)}{\widehat{var}(\hat{\theta}_w)}$$

- If $Inefficiency_0$ for $\hat{\theta}_w$ is unacceptably large then consider adding variables used in construction of the sample weights (sample design variables).
- These variables should not lessen the interpretability of θ , e.g., in a regression θ is the coefficient for an independent variable x and some of the added variables are **mediators** for the association of x to y .
- Do unweighted regression estimation with added sample design variables as (main effects) covariates, i.e., $\hat{\theta}_u$, and compare to weighted estimation for $\hat{\theta}_w$ with same added design variables. You can test if the estimated $\hat{\theta}_u - \hat{\theta}_w$ is significant (this test has low power). Substantive evaluation of the difference is often needed. If difference is large then use weighted estimation. Can consider truncating weights to improve efficiency of weighted analyses (Korn and Graubard 1995)

Statistical Models

- The **Normal distribution** is a model we often use in linear regression to describe features of population
 - Mathematical representation of reality
 - Do design-based methods use normal theory?
 - There is argument about this (Hansen et al., 1983, on Canvas)
 - **Design-Based Methods** : Use large samples and rely on convergence of the sample distribution to normality across hypothetical repeated samples

Bayes Models

- Wikipedia: “**Bayesian probability** is an [interpretation of the concept of probability](#), in which, instead of [frequency](#) or [propensity](#) of some phenomenon, probability is interpreted as reasonable expectation representing (incorporating) a state of knowledge or as quantification of a personal belief (subjectivity).”
- This in contrast to long run expectations defining the design-based (or frequentist) approach.
- Bayesian approach offers ways to build models and perform estimation and inference for complicated problems where frequentist are difficult and not obvious.
 - Hierarchical modeling as can be used small area estimation.
 - Missing data problems, e.g., multiple imputation

Bayes and Subjectivity

- Some Bayesian statisticians make a strong subjectivity assumption: probability is “quantification of a personal belief” as reflected in the extent of informativeness of the chosen prior distribution of the parameters to be estimated or inferred.
- Other Bayesians argue that prior knowledge is always employed to some extent, e.g., substantive knowledge could restrict the range of possible values the parameters can be. The Bayesian perspective is way to formalize this prior knowledge.
 - Prior distribution is explicit and needs careful justification
- May disagree about the nature of the prior:
 - Uninformative prior? (sometimes call a “diffuse prior” or “flat prior”)
 - Reflects a lack of knowledge of the parameters? What is a lack of knowledge?
 - Empirical Bayesian (construct priors based on (previous) data analyses)

Bayes Theorem

- Developed in the 18th Century
- For an event A and event B then the conditional probability of occurrence of event A given the occurrence of event B is written as $P(A|B) = \frac{P(A \cap B)}{P(B)}$

- Bayes Theorem says:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \text{ (because } P(B|A) = \frac{P(A \cap B)}{P(A)} \text{)}$$

- One application of Bayes Theorem: The probability that a patient has a disease given that they tested positive for that disease can be found using the probability that the test yields a positive result when the disease is present.

Application of Bayes Theorem

- An example:

- COVID testing (made up)
- Assuming 5% have COVID
- COVID test has a true positive rate of 90% (If have COVID, 90% test positive)
- COVID test has true negative rate of 80% (If not have COVID, 80% are false positive)
- What is the Positive Predictive Value (PPV)? True Positives/Tested Positives

- $PPV = P(COVID | Positive Test) = \frac{P(Positive Test | COVID)P(COVID)}{P(Positive Test)}$

	COVID	Not COVID	Total
Positive Test			
Negative Test			
Total			1000

Bayes Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Application of Bayes Theorem

- An example:
 - COVID testing (made up)
 - Assuming 5% have COVID
 - COVID test has a true positive rate of 90% (If have COVID, 90% test positive)
 - COVID test has true negative rate of 80% (If not have COVID, 80% are false positive)
 - What is the Positive Predictive Value (PPV)? True Positives/Tested Positives
- $PPV = P(COVID | Positive Test) = \frac{P(Positive Test | COVID)P(COVID)}{P(Positive Test)}$

	COVID	Not COVID	Total
Positive Test			
Negative Test			
Total	50		1000

Bayes Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Application of Bayes Theorem

- An example:
 - COVID testing (made up)
 - Assuming 5% have COVID
 - COVID test has a true positive rate of 90% (If have COVID, 90% test positive)
 - COVID test has true negative rate of 80% (If not have COVID, 80% are false positive)
 - What is the Positive Predictive Value (PPV)? True Positives/Tested Positives
- $PPV = P(COVID | Positive Test) = \frac{P(Positive Test | COVID)P(COVID)}{P(Positive Test)}$

	COVID	Not COVID	Total
Positive Test	45		
Negative Test	5		
Total	50		1000

Bayes Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Application of Bayes Theorem

- An example:
 - COVID testing (made up)
 - Assuming 5% have COVID
 - COVID test has a true positive rate of 90% (If have COVID, 90% test positive)
 - COVID test has true negative rate of 80% (If not have COVID, 80% are false positive)
 - What is the Positive Predictive Value (PPV)? True Positives/Tested Positives
- $PPV = P(COVID | Positive Test) = \frac{P(Positive Test | COVID)P(COVID)}{P(Positive Test)}$

	COVID	Not COVID	Total
Positive Test	45	190	
Negative Test	5	760	
Total	50	950	1000

Bayes Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Application of Bayes Theorem

- An example:
 - COVID testing (made up)
 - Assuming 5% have COVID
 - COVID test has a true positive rate of 90% (If have COVID, 90% test positive)
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- $PPV = P(COVID | Positive Test) = \frac{P(Positive Test | COVID)P(COVID)}{P(Positive Test)}$

	COVID	Not COVID	Total
Positive Test	45	190	235
Negative Test	5	760	765
Total	50	950	1000

Bayes Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Application of Bayes Theorem

$$\begin{aligned} \bullet \text{ } PPV &= P(COVID|Positive \text{ Test}) = \frac{P(Positive \text{ Test}|COVID)P(COVID)}{P(Positive \text{ Test})} \\ &= \frac{\frac{45}{50} \times \frac{50}{1000}}{\frac{235}{1000}} = \frac{45}{235} = 0.19 \end{aligned}$$

	COVID	Not COVID	Total
Positive Test	45	190	235
Negative Test	5	760	765
Total	50	950	1000

Bayes Theorem

- Start with selecting $P(\theta)$ the **Prior Distribution** of θ (based on prior belief) and the observed sample data $y = \{y_1, y_2, \dots, y_n\}$ then apply Bayes Theorem to obtain $P(\theta|y)$ **Posterior Distribution** of θ given observed data y :

$$P(\theta|y) = \frac{P(\theta)P(y|\theta)}{P(y)}$$
, θ are the parameters that are random quantities (unlike Design-Based and Frequentist approaches)

$P(\theta|y)$ is an **update** of prior distribution of θ due to y

$P(y|\theta) =$ **Likelihood** (of observed data, given θ), which is a model/distribution that is chosen

$P(y)$ called the **marginal distribution** of observed data y is obtained from $P(\theta)$ and $P(y|\theta)$ (considered a normalizing constant of the posterior distribution, i.e., it makes sure $P(\theta|y)$ sums/integrates to 1), dropping it gives:

$$P(\theta|y) \propto P(\theta)P(y|\theta), \text{ where } \propto \text{ is symbol for proportional}$$

Bayesian Analysis: Discrete Prior Distribution Example

- Study: Researchers screen for vision problems among children. Sample of $n=14$ children found 2 with vision problems.
- Want inference about the probability of having vision problems, θ .

- The likelihood $P(y|\theta)$ is chosen to be the binomial distribution

$$P(y|\theta) = \binom{14}{y} \theta^y (1 - \theta)^{14-y} \text{ e.g., when } P(y = 2|\theta = 0.12) = \binom{14}{2} 0.12^2 (0.88)^{12} = 0.2827$$

- We assume the only **possible** values of $\theta \in \{.10, .12, .14, .16, .18, .20\}$ and researchers have prior beliefs about their probabilities. Table below gives computation for

θ	$P(\theta)$	$P(y \theta)$	$P(\theta) \times P(y \theta)$	$P(y)$
0.10	0.10	0.257	0.0257	0.103
0.12	0.15	0.2827	0.0424	0.170
0.14	0.20	0.2920	0.0584	0.233
0.16	0.20	0.2875	0.0576	0.230
0.18	0.15	0.2725	0.0410	0.164
0.20	0.10	0.2501	0.0250	0.100
Total	1.00		0.2501	1.000

Note that 5th column is Posterior Distribution

$$P(\theta|y) = \frac{P(\theta) \times P(y|\theta)}{P(y)},$$

where $P(y) = \sum_{\theta} P(\theta) \times P(y|\theta)$

= 0.2501 Total in 4th column

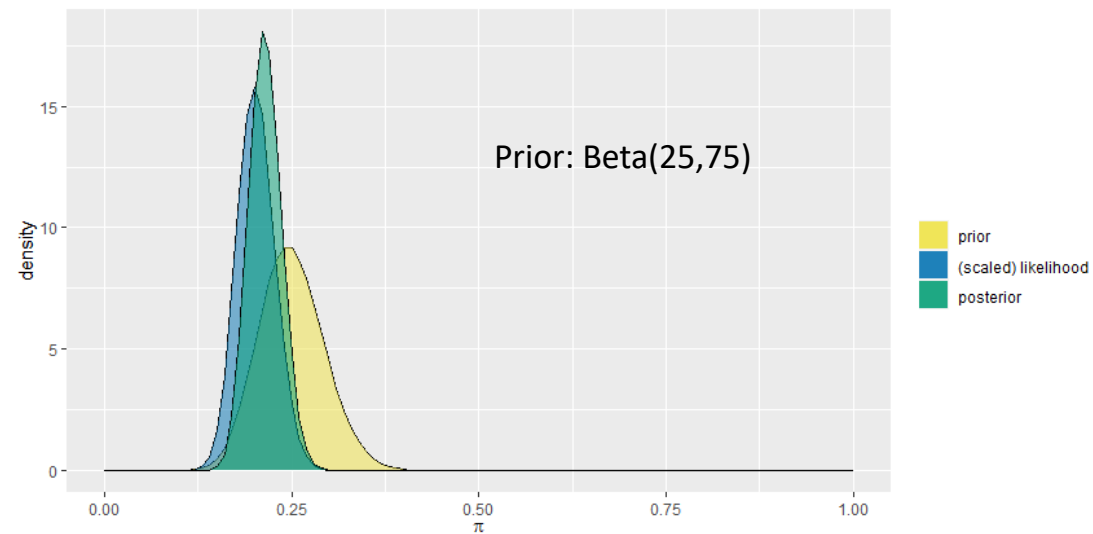
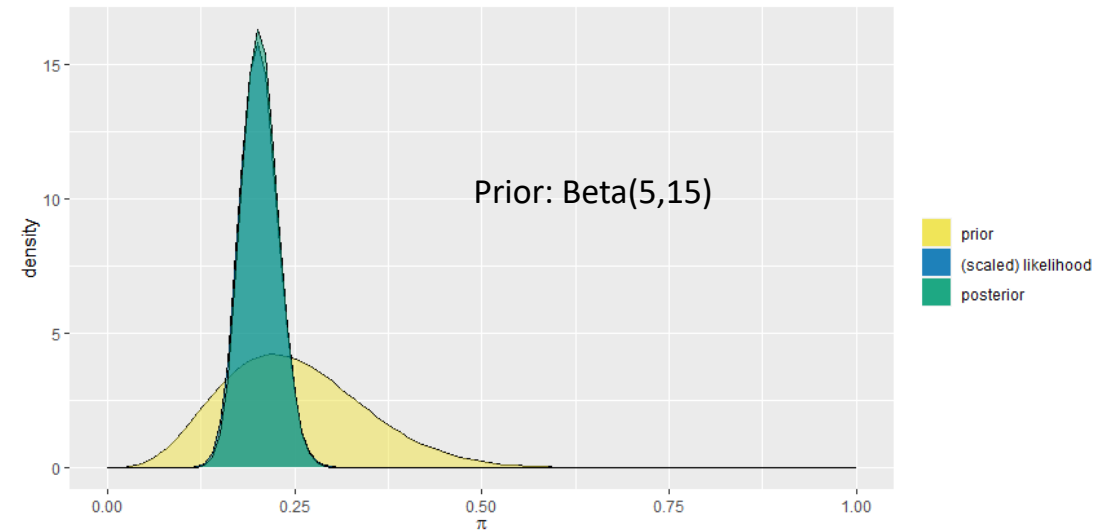
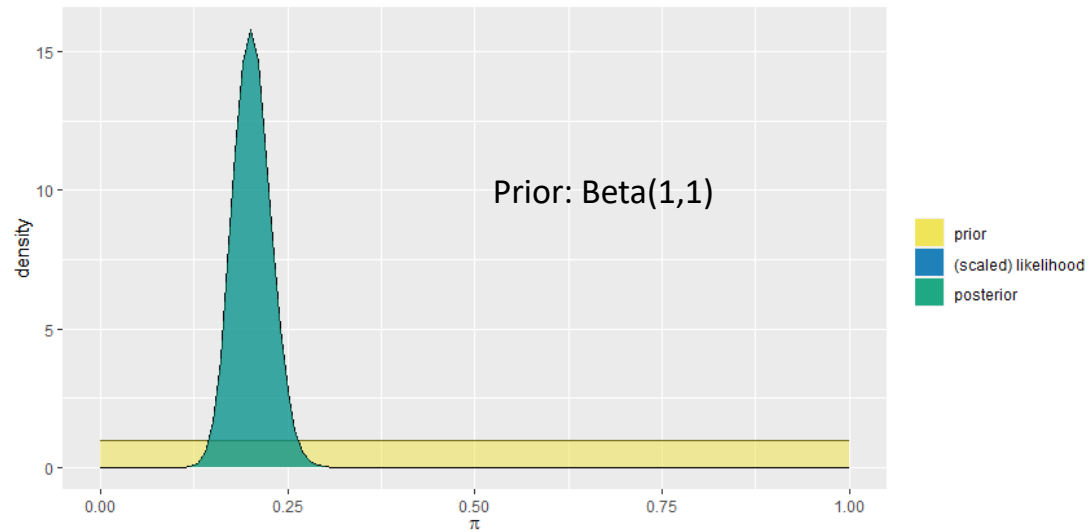
Bayesian Analysis: Continuous Prior Distribution Example

- **Beta-Binomial Model:** Want to make inference about a parameter, θ , that is the probability of a binary event across trials (e.g., poor vision among children, $n=14$).
- **Prior:** Specify a (subjective) **continuous prior** distribution for θ , a $Beta(a, b)$ distribution with density:

$$f(\theta, a, b) = \frac{\theta^{a-1}(1-\theta)^{b-1}}{B(a, b)}, 0 \leq \theta \leq 1, B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$$

- If the $P(\theta)$ is $Beta(1, 1)$, then it is a non-informative prior, i.e., a flat prior equal to 1, with expected value of θ of $\frac{a}{a+b} = 0.5$
 - Bayes is **subjective**: you choose the prior you want
- **Likelihood** $P(y|\theta)$ is Binomial(n, θ), where y is the number of successes
- **Posterior**, from Bayes Theorem: $P(\theta|y) \sim Beta(y + 1, n - y + 1)$
 - You **update the prior**, given new information!

Bayesian Analysis: Example



Likelihood: Binomial(250, θ)

Steps in Bayesian Data Analysis

- Set up a full probability model (subjective part)
 - A joint distribution for all observable and unobservable quantities
i.e., $P(\theta) \times P(y|\theta)$
- Condition on observed data
 - Given prior information and observed data, calculate a posterior distribution $P(\theta|y)$ (either for a **parameter** or **observations not sampled**)
- Evaluate model fit
 - How sensitive are conclusions of you inference to the choice of models (for prior and likelihood)?

Gelman et al. Bayesian Data Analysis (2013)

Broadband Internet Example

- Goal is to evaluate the **proportion** of the population that has **broadband internet**
- We can't visit every housing unit to measure this indicator
- Setting this up in a Bayesian fashion, our focus is now on determining a posterior distribution of the parameter predictions of values of the variable of interest for observations not sampled, i.e., **unobservable quantities (Bayesian Finite Population Inference)**

Broadband Internet Example (Cont.)

N = Finite Population Size

$$Y_i = \begin{cases} 1, & \text{if household } i \text{ has a broadband} \\ 0, & \text{otherwise} \end{cases}$$

$Q(Y) = \sum_{i=1}^N Y_i / N$ Proportion of households with broadband and is
the Finite Population parameter (this is the parameter)

Question of Interest: What is $P(Q(Y) \geq 0.3)$?

- We want to estimate the probability that $Q(Y)$ exceeds 0.3, i.e., meets criterion of “high internet penetration”
- Conduct a small-scale survey to answer the question of interest
- **Note: this question makes sense under Bayes paradigm as $Q(Y)$ is random (Under frequentist inference, $Q(Y)$ is assumed to be fixed but unknown. A frequentist approach could be to test the null $H_0 Q(Y) < 0.3$ vs. alternative $H_A Q(Y) \geq 0.3$)**

General Setup Used for Survey Data Analysis

- Model for $I = (I_1, I_2, \dots, I_N)$ for the Sample Design
- Model for $Y = (Y_1, Y_2, \dots, Y_N)$
- Frame/Sample Design Variables: Z
- Joint distribution $P(Y, I|Z)$
- Observed Data (Y_{inc}, I, Z) , Y_{inc} are Y 's included in the sample
- Missing or Unobserved Data, Y_{exc}
- Inference: $\Pr(Y_{exc}|Y_{inc}, I, Z)$
- We will use the Bayesian approach to generate **posterior predictions** of Y for non-sampled units, Y_{exc}

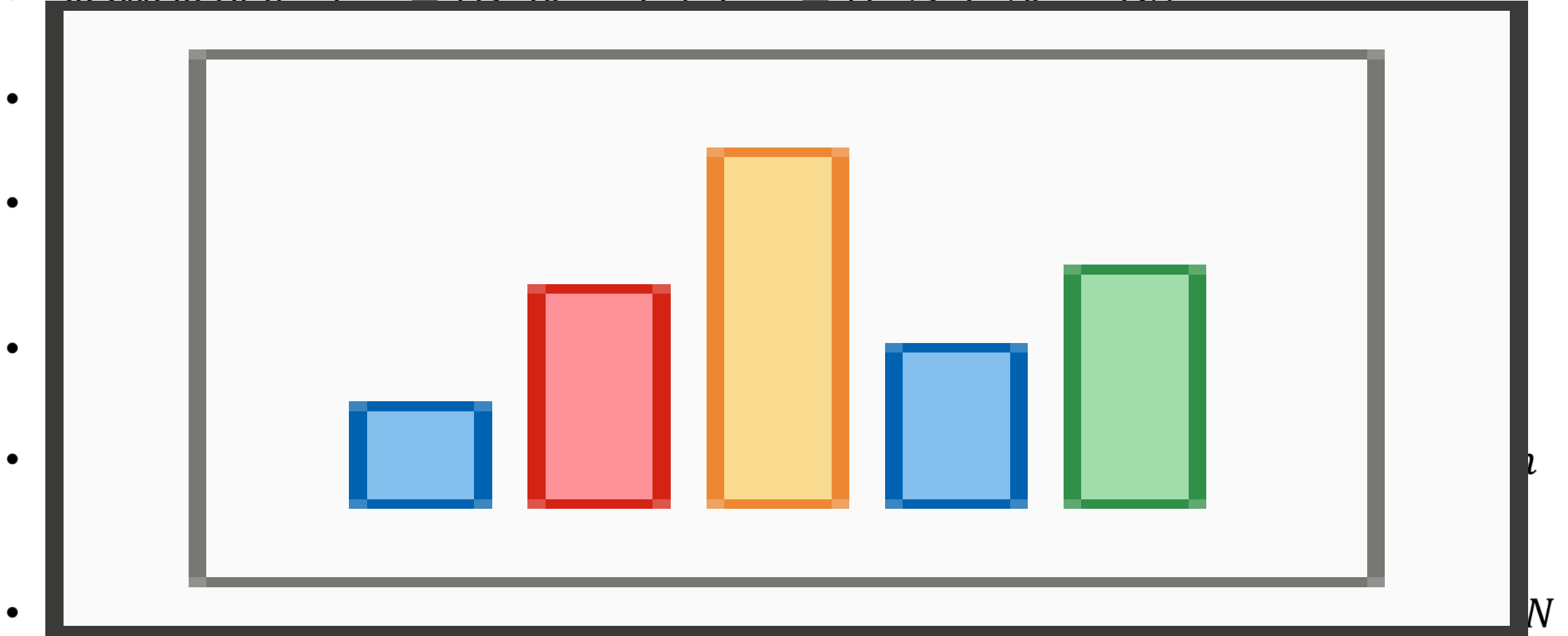
I	Z	Y
1		Y_{inc}
1		
1		
0		Y_{exc}
0		
0		
0		
0		

Under Simple Random Sampling Without Replacement

- No Z for SRSWOR
- Ignorable probability sampling, i.e., $Pr(Y, I) = Pr(Y)Pr(I)$

Broadband Example

- SRSWOR of n $Y_i = \{Y_1, Y_2, \dots, Y_n\}$ $Y = \{Y_1, Y_2, \dots, Y_N\}$



Binomial Example

The posterior distribution $p(\theta|x)$ is:

$$p(\theta|x) = \frac{p(x|\theta)\pi(\theta)}{\int p(x|\theta)\pi(\theta)d\theta} \propto p(x|\theta)\pi(\theta)$$

$$p(\theta|x) = \frac{\binom{n}{x}\theta^x(1-\theta)^{n-x} \times 1}{\int_0^1 \binom{n}{x}\theta^x(1-\theta)^{n-x}d\theta}, \text{ since } \pi(\theta) = 1, 0 \leq \theta \leq 1$$

The $p(\theta|x)$ is a $Beta(x+1, n-x+1)$ distribution.

$Q(Y) = (x + \sum_{i=n+1}^N Y_i)/N$, is the estimand

$$(\sum_{i=n+1}^N Y_i | \theta, x) \sim \text{Bin}(N-n, \theta)$$

The general idea use simulation to get an approximate distribution of $Q(Y)$, **given a posterior distribution** of θ :

1. Randomly draw a value of θ from the **posterior distribution** $p(\theta|x)$, $Beta(x+1, n-x+1)$, (which is informed by the observed x and n).
2. Given that draw of θ , randomly draw a value of the sum of the Y values for the **non-selected units** from the stated Binomial distribution, $\text{Bin}(N-n, \theta)$, and compute the estimand $Q(Y)$
3. Repeat steps 1 and 2 many times, and look at the distribution of the “drawn” values of $Q(Y)$ for answering the research question, e.g., $P(Q(Y) \geq 0.3)$?

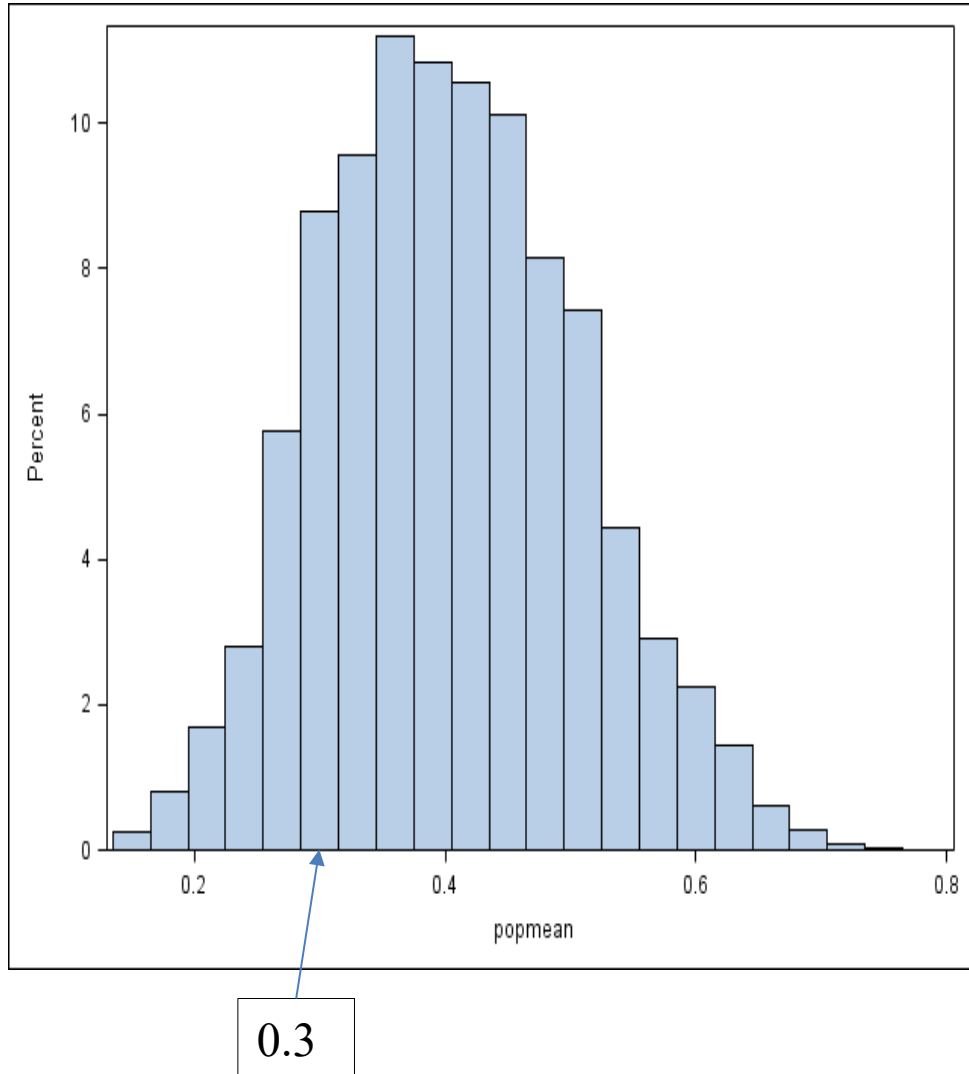
Inference for Population Parameter θ

- Recall the Question of Interest: What is $P(Q(Y) \geq 0.3)$?, $Q(Y) = \bar{Y}_N$
- Since as $N \rightarrow \infty, \bar{Y}_N \rightarrow \theta$ then $\Pr(\bar{Y}_N \geq 0.3|x) \approx \Pr(\theta \geq 0.3|x)$
- Compute $\Pr(\theta \geq 0.3|x)$ using **cumulative posterior distribution**, i.e., the cumulative distribution function (CDF) of the beta distribution, $Beta(x + 1, n - x + 1)$, which is a standard function in most software, such as R.

Numerical Example

- Finite population of $N = 270$ households
- SRS sample of size $n = 20$
- Suppose $x = 8$ out of sample of 20 had broadband
- *Simulation Steps:*
 - (1) Randomly draw θ from posterior, $\theta \sim \text{Beta}(9, 13)$, from $\text{Beta}(x + 1, n - x + 1)$, $x = 8$ and $n = 20$.
 - (2) Randomly draw the sum $X_{N-n} \sim \text{Bin}(250, \theta)$ given the draw of θ in Step (1)
 - (3) Compute a "draw" of the estimand, $\bar{Y}_N = (x + X_{N-n})/N$
 - (4) Repeat many times (say 1,000) to get a simulated posterior distribution of $\bar{Y}_N$ then use that simulated distribution to obtain a credible interval of $\bar{Y}_N$.

Histogram of 2,500 Draws



Proportion of draws of $Q(Y) = \bar{Y}_N$ exceeding 0.3: 0.84 (the probability that $\bar{Y}_N$ exceeds 0.3 in this area is 0.84)

Posterior mean: 0.4051 (point estimate of $\bar{Y}_N$)

Posterior standard deviation: 0.1007
(measure of variance in estimate)

Use Normal Approximation to obtain 95% credible interval for population $\bar{Y}_N$:

$$(0.2077, 0.6025) = 0.4051 \pm 1.96 \times 0.1007$$

Point Estimates in Bayesian Analysis

- The point estimate is often a single summary “best” value for the unknown estimand $Q(Y)$
- Some choices are the mean, mode or the median of the **posterior distribution** of $Q(Y)$
- For symmetrical distributions, an intuitive choice is the center of symmetry
- For asymmetrical distributions, the choice less clear.

Credible Interval Estimation

- A credible interval is used to summarize Bayesian inference, which is similar to confidence interval used in frequentist and design-based inference
- There are different methods to obtain credible intervals :
 - (1) We just used normal distribution approximation for broadband example (Slide 58) to get an interval centering around the mean of the posterior distribution, i.e., $\text{mean} \pm 1.96 \times \text{std}$ of the posterior distribution.
 - (2) Fix the coverage rate $1 - \alpha$, e.g., 95%, in advance and determine the *highest posterior density (HPD)* region, C , to include most likely values of $Q(Y)$ totaling 95% of the posterior probability. For symmetric unimodal posterior distributions, the $(1 - \alpha)$ HPD interval is $(q_{\alpha/2}, q_{(1-\alpha)/2})$ where $\Pr(Q(Y) \leq q_{\alpha/2}) = \alpha/2$. For example, use the 2.5 and 97.5 percentiles of the simulated draws of the estimand, $Q(Y)$, to form a 95% credible interval.
- In the Broadband example with $Y_i \sim \text{Bernoulli}(\theta)$, the beta density for the posterior can be used to determine the credible interval of $Q(Y)$.

Goal of Bayesian Approach for Complex Surveys

Derive (or simulate) the **Posterior (predictive) distribution**, conditional on the design variables Z, I :

$$P(Y_{exc} | Y_{inc}, Z, I)$$

Two-stage Construction (draws of parameters from posterior, then draws of values for excluded cases given drawn values of parameters, as in the broadband example):

$$\pi(\theta | Y_{inc}, Z, I)$$

$$P(Y_{exc} | \theta, Z)$$

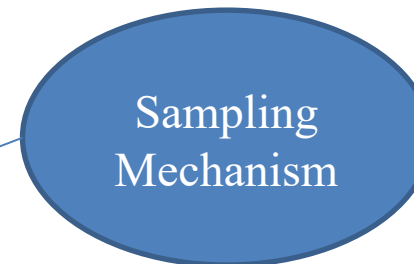
General Setup: Complex Samples

I : Inclusion indicator

Y : Survey Variables

Z : Design Variables

Model : $\Pr(Y, I \mid Z) = \Pr(Y \mid Z) \Pr(I \mid Y, Z)$



$$\Pr(Y \mid Z) = \int \Pr(Y \mid Z, \theta) \pi(\theta \mid Z) d\theta$$

Design variables Z include Weights, Clustering, Stratification.

There may be additional auxiliary variables not part of the design but predictive of Y and available for all subjects in the population

Observed Data : (Y_{inc}, I, Z)